A thirteen-variable consistency equation

Positive integer unknowns; exact degree 36; an 87-operation evaluator.

The only unknowns are the thirteen distinct positive integers

$$J, F, \alpha, f, g, T, X, Y, \xi, \eta, \zeta, \upsilon, \gamma \in \mathbb{Z}_{>0}.$$

All other symbols below are polynomial abbreviations:

$$\begin{gathered}
b = \beta J, \quad q = 1 + b, \quad C = q - F - \alpha, \quad U = Yq^2, \\
E = XY, \quad k = \eta + \zeta, \quad \ell = EYk, \quad c = \eta + Yk, \\
a = E + Y, \quad H = 3 + 4a, \quad \Delta = H + a^2, \quad L = X + ac, \\
M = q^3 b, \quad W = C(X + Kq) - Fq, \\
R = [\alpha + (q - F)b](q^2 - 1) + J(\mu + \omega q), \\
Z = M(\ell + k + \xi - kW) + Yb - Wq^3, \\
D = UL + H(U\gamma + Eq + R + 1 - k), \\
I = c^2 g, \quad Q = \Delta(f^2 - 1), \quad V = f^2(Tc - R) - c, \\
\mathcal{P} = Z^2 - M^2[1 + \ell(\ell + k)], \\
\mathcal{D} = D^2 - U^2(1 + \Delta c^2), \\
\mathcal{Q} = \upsilon^2 - Q(\upsilon^2 - V^2) - 1, \quad \mathcal{R} = I^2 - Q.
\end{gathered}$$

The single Diophantine equation is

$$\boxed{\mathcal{P}^2 + \mathcal{D}^2 + \mathcal{Q}^2 + \mathcal{R}^2 = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho = 2^{47}, \quad B = \rho^{2011}, \quad \beta = B - 1, \\
G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1005}.
\end{gathered}$$

$$\begin{aligned}
\mu &= \beta - G_{951}(\rho) \\
&\quad - \rho^{960} G_{26}(\rho) - \rho^{1004}, \\
\omega &= \beta + 8G_{917}(\rho) - 4 + (2^{45} - 4)\Lambda \\
&\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1004}, \\
K &= \rho^{18} + \rho^{1004} + \rho^{2020} \\
&\quad + 2\Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{gathered}
\chi_i = 1 + 2^{10}(1 + w_i + 2^8 n_i \\
+ 2^{17} p_i + 2^{25} s_i) \quad (0 \le i \le 941), \\
\pi(0) = 0, \quad \pi(i) = 1 + \sum\nolimits_{j=1}^{941} \left\lfloor \frac{2^{44} - 1 + \chi_j - \chi_i}{2^{44}} \right\rfloor \quad (1 \le i \le 941).
\end{gathered}$$

$$\begin{aligned}
\Phi(x) = \ & \sum\nolimits_{i=0}^{941} \chi_i x^{\pi(i)} \\
& + 2^{10} x^{969}(x^9 + 2^{17}) G_8(2x) \\
& + 2^{18} x^{942}(x^{18} + 2^{17}) G_9(2x).
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$\begin{gathered}
h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\}, \\
r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j + 1$ $(j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$	$(dr_{d,q}, b, (1 - d) r_{d,q}, h(q, b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
\mathcal{N} = \mathcal{A} \bmod 2^{636}, \quad \mathcal{T} = \left\lfloor \mathcal{A}/2^{636} \right\rfloor, \\
\nu_j = D_2(\mathcal{N}, j), \quad \kappa_j = (\sigma_j + 2)^{1 - \nu_j} \quad (0 \le j < 430), \\
\sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \quad \Pi_j = \prod_{k=0}^{j-1} \kappa_k \quad (0 \le j \le 430), \\
m_q = D_2(\mathcal{N}, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14 + m_q(D_{16}(\lfloor \mathcal{T}/\Pi_{430} \rfloor, \tau_q) - 14), \\
u_j = (\sigma_{j+1} - (\lfloor \mathcal{T}/\Pi_j \rfloor \bmod \kappa_j)) \bmod 216, \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

```
12889260410359561746736540743515845134244260640012702033212621
95398074180784058571518488282119861023093355389406939414042892
97455773992193784795705001618777025966369137954859764532832722
75411843410809038713502633204617333246733616940964578272360901
56186216508504953649192752154519268024509952082950238865468307
22999632482769253980274091217460707829023970817244790340704449
09085581271192071656170732004937337082171141082486384269072884
72778087419663417942892520175630649792329265834490253035577533
96449689808375788457488971017043689875433162972390177936123032
89025657757489526342497438772045714099661532481123050235898973
27443609903800541199351405985754278307227777584454508422717998
33411356440933534607489268032843 7
```

The fixed integer $\mathcal{A}$ above is written in base 10. Lean proves that this equation and the 86-operation equation have the same positive solutions in every model used by the arithmetic proof. The construction of an intuitionistic PRA proof of the consistency equivalence is in progress. The exact degree is certified by integer-polynomial calculations. The 87-operation count is for the unshifted circuit; neither count is a minimality claim.

Exact 87-operation circuit

Each numbered line defines one gate over the integers. The inputs are the thirteen unknowns and the four fixed integers above. The ASCII names `alpha`, `xi`, `eta`, `zeta`, `upsilon`, `gamma`, `beta`, `mu`, `omega`, `ell`, `Delta` refer to their displayed counterparts; `g` is the unknown g. The last gate is zero exactly when the displayed equation holds. Intermediate gate names below are local abbreviations, not additional unknowns.

```
1.  b = J * beta
2.  q = 1 + b
3.  q2 = q * q
4.  q3 = q * q2
5.  U = Y * q2
6.  qF = q - F
7.  C = qF - alpha
8.  E = X * Y
9.  k = eta + zeta
10.  kY = Y * k
11.  c = eta + kY
12.  c2 = c * c
13.  a = E + Y
14.  g15 = 4 * a
15.  H = 3 + g15
16.  g17 = a * a
17.  Delta = H + g17
18.  g19 = a * c
19.  L = X + g19
20.  ell = E * kY
21.  M = q3 * b
22.  g25 = K * q
23.  g26 = X + g25
24.  g27 = C * g26
25.  g28 = F * q
26.  W = g27 - g28
27.  g30 = ell + k
28.  g31 = g30 + xi
29.  root_skew = k * W
30.  root_inner = g31 - root_skew
31.  root_scaled = M * root_inner
32.  g33 = Y * b
33.  root_btq3 = W * q3
34.  root_sum = root_scaled + g33
35.  Z = root_sum - root_btq3
36.  g38 = Z * Z
37.  g39 = M * M
38.  g40 = ell * g30
39.  g41 = 1 + g40
40.  g42 = g39 * g41
41.  A0 = g38 - g42
42.  g44 = qF * b
43.  rankseed = alpha + g44
44.  g46 = q2 - 1
45.  g47 = rankseed * g46
46.  g48 = omega * q
47.  g49 = mu + g48
48.  g50 = J * g49
49.  R = g47 + g50
50.  g52 = k - R
51.  r0 = g52 - 1
52.  g54 = U * gamma
53.  g55 = E * q
54.  g56 = g54 + g55
55.  shift = g56 - r0
56.  g58 = U * L
57.  g59 = H * shift
58.  D = g58 + g59
59.  g61 = D * D
60.  g62 = U * U
61.  g63 = Delta * c2
62.  g64 = 1 + g63
63.  g65 = g62 * g64
64.  A1 = g61 - g65
65.  f2 = f * f
66.  rho = f2 - 1
67.  Q = Delta * rho
68.  I = c2 * g
69.  g71 = I * I
70.  g72 = g71 - Q
71.  g74 = T * c
72.  g75 = g74 - R
73.  g76 = f2 * g75
74.  V = g76 - c
75.  y2 = upsilon * upsilon
76.  g80 = V * V
77.  g81 = y2 - g80
78.  g82 = Q * g81
79.  Fa = y2 - g82
80.  auxResidual = Fa - 1
81.  firstSquare = A0 * A0
82.  mainSquare = A1 * A1
83.  auxSquare = auxResidual * auxResidual
84.  strongSquare = g72 * g72
85.  firstMain = firstSquare + mainSquare
86.  firstMainAux = firstMain + auxSquare
87.  sumSquares = firstMainAux + strongSquare
```